Chapter 13 & 15: Production Costs & Competitive Markets

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Discussion section 1

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Outline

Today we will talk about:

- Costs of Production!
- Firms in Competitive Markets (Time Permitting)

Costs of Production

A firm aims to maximize their profit:

$$\text{Profit} = \text{total revenue} - \text{total cost}$$

- total revenue: $TR = P \times Q$
- total cost is the sum of all **implicit** and **explicit** costs of inputs to production
  - **Implicit**: input costs which *do not* require monetary outlay from the firm
  - **Explicit**: input costs which *do* require a monetary outlay from the firm

Types of Profit

There are two types of profit we can consider:

Economic Profit:

total revenue less implicit *and* explicit costs

Accounting

total revenue less explicit costs *only*

Costs

Types of costs:

- Fixed costs: costs that do not vary with Q
- Variable costs: costs that vary with Q

Measures of costs:

- Average total cost (ATC): $ATC = TC/Q$
- Average fixed cost (AFC): $AFC = FC/Q$
- Average variable cost (AVC): $AVC = VC/Q$
- Marginal Cost: $MC = \Delta TC/\Delta Q$

Example

Consider a firm which produces pizzas:

A worker costs \$100 and the firm has fixed costs of \$200.

The following table shows the number of pizzas produced given varying numbers of workers. Let's add information for the marginal product, total cost, average total cost, and marginal cost.

Workers	Output
0	0
1	20
2	45
3	80
4	100
5	110

What do costs look like?

- We tend to assume (like in previous example) that the marginal product of labor is decreasing... Why?
- Average fixed cost falls as output rises because the same fixed cost is spread over more units.
- Average variable cost typically falls at first and then rises: initially, workers can specialize and use fixed inputs more effectively, but eventually diminishing marginal product sets in.
- Marginal cost also often has a U-shape because when marginal product is rising, each additional unit is relatively cheap to produce, but when marginal product falls, each additional unit becomes more costly.
- Average total cost is the sum of average fixed cost and average variable cost, so it often has a U-shape: it falls at first, then rises once the increase in AVC outweighs the continued decline in AFC.

What do costs look like?

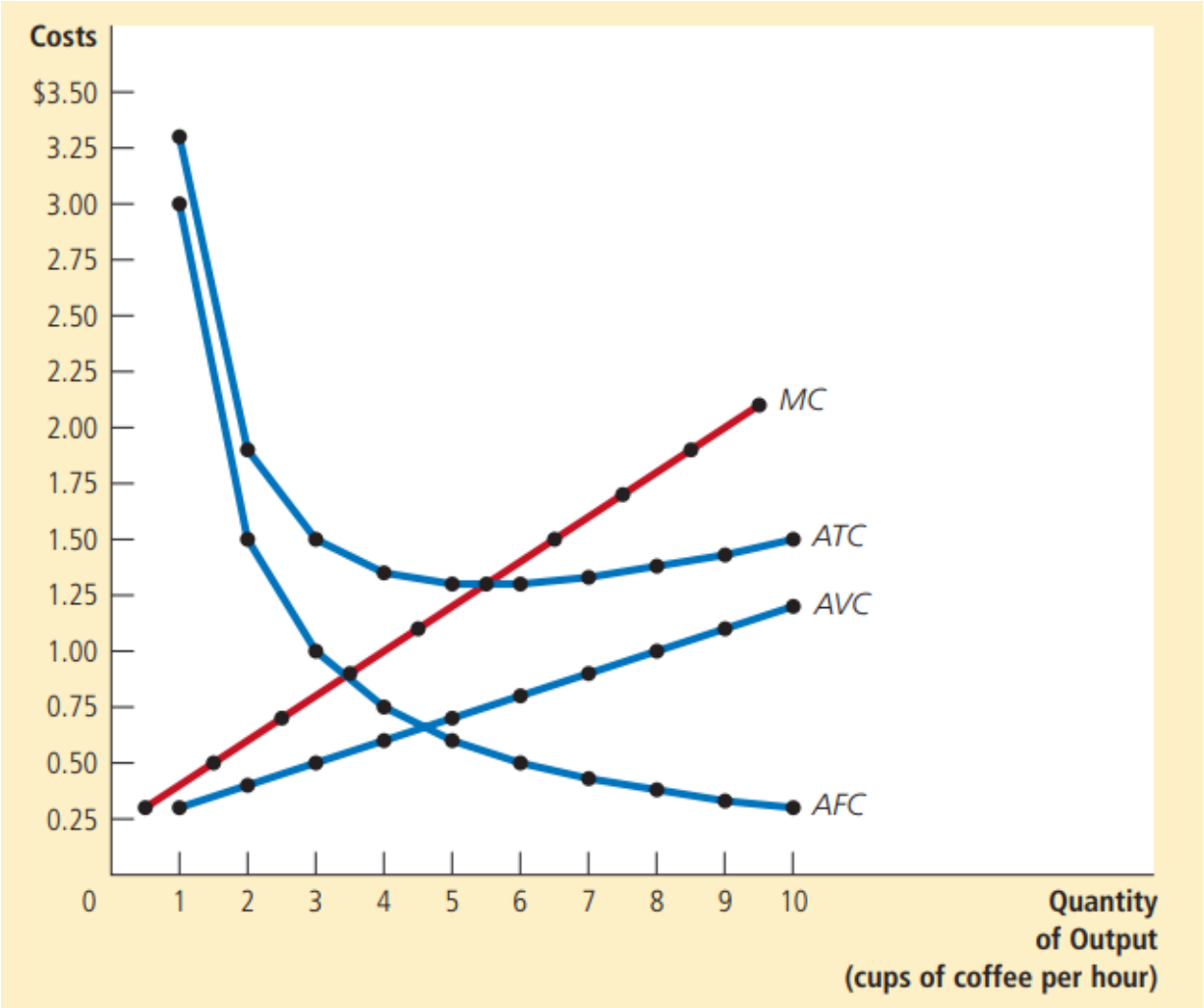

Figure 1: Relationship between costs

More Realistic Cost curves

Let's remember what our cost curves look like:

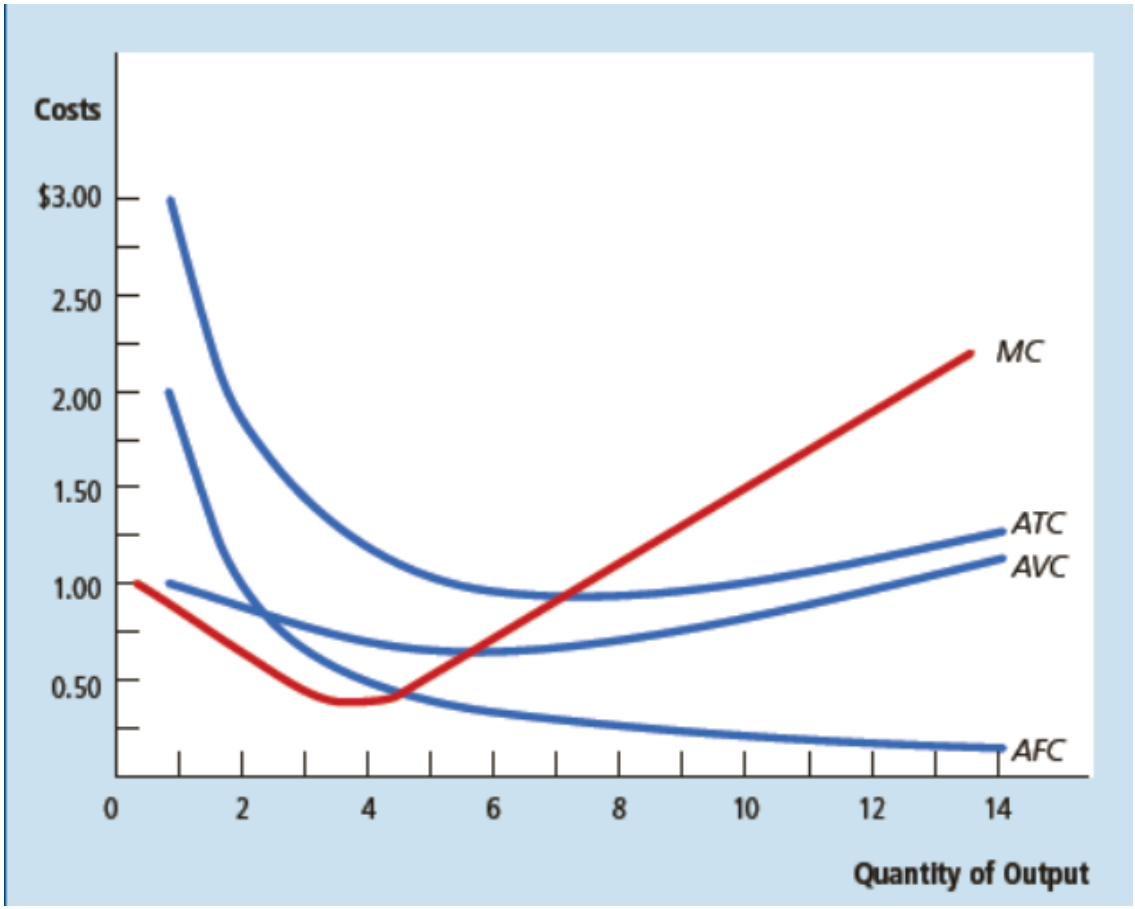

Figure 2: Realistic Cost Curves

A Final Piece of Intuition

The marginal cost curve intersects average total cost at the minimum of ATC: if MC is below ATC, producing one more unit pulls the average down; if MC is above ATC, producing one more unit pushes the average up.

Economies and Diseconomies of Scale

We can define the scale of production in the following 3 ways:

1. Economies of Scale: Long run ATC declines as Q increases
2. Diseconomies of Scale: Long run ATC increases as Q increases
3. Constant returns to scale: Long run ATC stays the same as Q changes

Competitive markets

Let's recall what we mean by "competitive market":

- Many buyers and sellers $\implies$ No single buyer or seller has influence over price
- The goods/services being sold are homogeneous
- Firms can freely enter/exit the market *(this is new!)*

What does this mean for firms?

- Firms are price takers
  - P remains constant for a firm at any level of Q $\implies$ $MR = P$

Competitive Markets

Firms in competitive markets want to maximize their profit.

Recall that profit is:

$$\pi = TR - TC$$

Total revenue (TR) is:

$$TR = P \times Q$$

We also consider average and marginal revenue:

- Average revenue: $AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P$
- Marginal revenue: $MR = \frac{\Delta TR}{\Delta Q}$ $\quad$ ($= P$ in a competitive market)

Optimal Choices in a competitive market

Firms will always maximize their profit when they choose to produce the Q where $MR = MC$!

- This is a direct result of the firms thinking on the margin:
  - If $MR > MC$ the firm should increase Q
  - If $MR < MC$ the firm should decrease Q
- **** The only Q where they do not want to make a change is the Q where $MR = MC$ ****

Because MC determines Q for any P (we look for Q such that $MR = MC$ and in a competitive market $MR = P$), the MC curve is the supply curve for a *competitive* firm

Let's fill in the following table:

Price	Quantity	TR	TC	π	MR	MC	Δπ
$6	0		$3		--	--	--
$6	1		$5				
$6	2		$8				
$6	3		$12				
$6	4		$17				
$6	5		$23				
$6	6		$30				

Firm Shutdown

Importantly, a firm does not want to operate at a loss (negative profit)

In the short run, a firm decides whether to remain open or shut down *for a period of time* (Not permanently!!!)

A firm should shutdown in the short run if:

$$P < AVC$$

- If the firm earns less per unit sold than the **variable cost** of producing that unit, they should shut down
- In the SR, there are fixed costs (rent, machines, etc.) that must be paid for whether or not the firms shuts down temporarily
  - These costs are called **sunk costs,** and we don't consider them in short term decisions

Firm Exit

Importantly, a firm does not want to operate at a loss (negative profit)

In the long run, a firm decides whether to remain in the market or exit the market *(Permanently!!!)*

A firm should exit the market in the long run if:

$$TR < TC \iff P < ATC$$

- If the firm earns less per unit sold than the *total cost* of producing that unit, they should exit the market
- We consider fixed costs in the long run because we can end our lease on the building/sell the machines in the long run
  - *These costs are not sunk in the long run*
- We can say that a firm should enter the market if the inequality is reversed

Supply curves

The supply curve of a competitive firm is the portion of their MC curve where they remain in the market, which is different in the short and long run...

In the short run

- Supply is 0 when $MC < AVC \quad (MC = MR = P)$
- Supply curve is the portion of MC curve which lies above the AVC curve

In the long run

- Supply is 0 when $MC < ATC$
- Supply curve is the portion of MC curve which lies above the ATC curve

How does shutdown/exit affect market equilibrium?

In the short run...

Firms may shutdown $\implies$ we may be pushed away from equilibrium

In the long run...

Firms may exit the market $\implies$ supply shifts left (less sellers)

- Drives prices up
- More firms enter the market because the price is higher
- Price drops back to the equilibrium level
- Firms that remain in the market must be making zero economic profit ($MC = MR$)
- Entry/Exit only stops when $ATC = MR = MC = P$

Graphing the market

Let's graph and calculate/shade the short run profit for a firm in the following two scenarios:

1. $MC > AVC$ and $MC > ATC$
2. $MC > AVC$ but $MC < ATC$

Now imagine the same firms in the long run... what does the long run profit look like on the graphs?